Ensemble Learning Based Quantum Text Classifiers

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Abstract. Quantum Natural Language Processing (QNLP) is a very young area of research, aimed at the design and implementation of NLP models that exploit certain quantum phenomena such as superposition, entanglement, and interference to perform language-related tasks on quantum hardware. To explore the structural relationships between quantum theory and natural languages Lambeq toolkit was created. This first high-level open-source Python toolkit for quantum natural language processing offers a fully automated quantum machine learning pipeline. Lambeq currently includes five compositional models that use varying degrees of syntactic information. While the previous studies focus on Discocat model, the aim of this study is to conducte an extensive evaluation of various classifiers based on different compositional models for text classification applications, focusing on the potential advantages of ensemble learning. The classifiers, including Spider, Cups, Stairs, Tree, and DisCoCat, were examined on two datasets: MC and RP. Performance evaluation was carried out using accuracy, precision, recall, and F1-score as the metrics. The results revealed notable variations in the performance of the classifiers across the datasets. Spider emerged as the top-performing classifier on the MC dataset, achieving remarkable scores of 100% in accuracy, precision, recall, and F1-score. However, on the RP dataset, Stairs outperformed the other classifiers with an accuracy of 68%, precision of 71%, recall of 68%, and F1-score of 69%. Furthermore, ensemble models using hard voting and soft voting techniques were constructed and evaluated. The ensemble models showcased improved performance compared to individual classifiers, with the soft voting ensemble achieving a score of 97% for all metrics on the MC dataset, and a precision of 72% and F1-score of 68% on the RP dataset. These findings highlight the potential benefits of ensemble learning in enhancing the overall performance of text classification tasks on both the MC and RP datasets using soft voting method, allowing for more accurate and reliable predictions.

Keywords: Ensemble Learning · Quantum machine learning · Quantum natural language processing · Quantum Text Classifier

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1 Introduction

Machine translation, question answering, text classification, sentiment analysis [10] and many other NLP applications have been considerably enhanced by recent developments in neural language models based on Transformers architecture. As an example, the BERT Large uses 24 layers of transformers block with a hidden size of 1024 and number of self-attention heads as 16 and has around 340M trainable parameter. Because more parameters and data are needed to train this type of models efficiently, there is a cost in terms of time, resources, and computing power. This results an increase in model complexity. Therefore, it is necessary to search for new ways to process natural language with less complexity.

However, quantum computing has attracted a lot of attention recently. The fundamental idea is to use quantum mechanics to solve computational problems, in order to following different paths of computation at the same time, quantum computers can make use of a qubit's superposition of two quantum states $\mid 0\rangle and \mid 1\rangle$. A significant advance in quantum computing was made by quantum algorithms such as Short perform calculations with lower complexity than conventional methods. Naturally, the field of machine learning benefits from the ideas that quantum computing has to offer which led to the emergence of a new sub-field called Quantum Machine Learning (QML). The need to expand the NLP field in new directions coincided with the development of quantum machine learning, which resulted in the emergence of so-called Quantum Natural Language Processing (QNLP), which represents today one of the best applications of quantum computing.

The following is how this paper is structured: Sect. 2 presents the main related work for QNLP. In Sect. 3 describes the data used in this paper and a comparative evaluation of baseline classifiers. In Sect. 4 the proposed ensemble schema is presented and the results are discussed. Finally in Sect. 5, we draw the main conclusions and discuss possible future works.

2 Related Work

The concept behind the implementation of natural language on quantum hardware is to create a *link between language meaning and grammatical structure and quantum states*. This connection is made using the Categorical Distributional Compositional (DisCoCat) model [3], where the meanings of words are vectors in vector spaces and words grammatical roles are types in a Pregroups then a tensor product of vector spaces paired with the Pregroup composition is used for the composition of (meaning, type) pairs. Authors in [11] demonstrate that a quantum algorithm for calculating phrase similarity yields a quadratic speedup over conventional approaches under specific conditions. However, it requires quantum random access memory (QRAM), which is costly and has yet to be implemented. As a result, [2] and [6] articles presents quantum algorithms that can potentially be implemented in current NISQ devices (Noisy intermediate-scale quantum).

These last papers assume that NLP is quantum native, in the sense that the exponentially massive vector space necessary to represent sentences can only be realized naturally and practically in quantum computers. Following that, various investigations in the literature have shown that quantum speedup is more likely to impact NLP tasks like question answering [7], machine translation [8] and classification [5]. In order to close the gap between theory and practice, eventually contributing to actual real-world QNLP implementations, Cambridge Quantum Computing present Lambeq, 'an open-source, modular, extensible high-level Python library, which provides the necessary tools for implementing a pipeline for experimental QNLP. At a high level, the library allows the conversion of any sentence to a quantum circuit, based on a given compositional model and certain parameterisation and choices of ansatze' [4]. This first high-level open-source Python toolkit for quantum natural language processing is still in progress to offer a fully automated quantum machine learning pipeline.

3 Methodology

The aim of this study is to investigate the potential of employing ensemble learning to improve prediction of quantum text classification tasks. As a result, the study was divided into two distinct phases, the first in which a comparative evaluation of five classifiers on two well-Known QNLP datasets. Then, an ensemble learning was designed and implemented using two different voting techniques: Hard and Soft Voting.

3.1 Dataset

Table 1. Description of MC and RelPron datasets.

Dataset	Number of sentences	Vocabulary	Classes
Meaning classification (MC)	130 sentences	17 words	IT or Food
Relative Pronoun (RP)	105 sentences	115 word	Subject or Object

This research uses the MC and RP datasets (See Table 1), a specific datasets for QNLP tasks proposed by [5]. MC dataset is plain-syntax generated from a 17 words vocabulary using a simple CFG that can refer to one of two possible topics, food or IT. RP dataset extracted from the original RelPron dataset [9] for predicting whether a noun phrase contains a subject-based or an object-based relative clause, with 115 words vocabulary. The motivation for using RP dataset, is that it requires some syntactic understanding from the model, making it a suitable option for evaluating lambeq's models. Furthermore, the large amount of the vocabulary and, as therefore, the sparsity of words make this task a significantly more challenging benchmark than the MC dataset.

3.2 Baseline Classifiers

In order to establish a foundation for comparison, we employed several baseline classifiers based on Lambeq's compositional models, which generate semantic representations of sentences by combining the semantic representations of the words within them. Currently, the Lambeq toolkit provide five compositional models that use different levels of syntactic information. As described in the official documentation of Lambeq library there are three levels of syntactic information [1]:

1. No syntactic-information: include Spider reader, a bag-of-word model which renders simple monoidal structures without any form of tagging.
2. Word-sequence information: where words are formed sequentially from left right. Cups and Stairs are the two models available.
3. Full syntax-information: derived from grammatical derivations BobCatParser and TreeReader are two examples of syntax-based lambeq models. The Distributional Compositional Categorical model introduced by [3] is implemented by BobCatParser. Tree reader, on the other hand, provides a CCG derivation that follows a biclosed form, which may be directly understood as a series of compositions without any explicit translation into a pregroup form.



Fig. 1. Full Quantum Pipeline

Figure 1 describes a process involving the conversion of a sentence into a quantum circuit in lambeq. Initially, a syntax tree is obtained for a sentence using the parser, and then it is converted into a string diagram, representing the relationships between words based on a compositional model of choice. The string diagram can be simplified or transformed using rewriting rules, making it suitable for implementation on a quantum processing unit. The resulting diagram is converted into a quantum circuit or a tensor network, depending on the experiment type. The output of this pipeline, whether a quantum circuit or tensor network, can be used for training.

In our study, we investigate the usage of quantum classifiers with different semantic representations; as a consequence, we nominate our quantum classifiers with the same name as the compositional model used to identify them, such as the Stairs, Spiders, Cups, Tree, and DisCoCat classifiers. These classifiers served as reference models against which we evaluated the performance of our Ensemble model. The baseline classifiers were selected based on their varying degrees of syntactic information, simplicity in deployment, and the absence of literature on their effectiveness (reserve for the DisCoCat model).

3.3 Comparative Evaluation

To ensure fair and consistent comparisons, all baseline classifiers were trained and evaluated using the same experimental setup and evaluation metrics. The Removecups function was used to simplify string diagrams, and the IQP ansatz was utilized to generate the corresponding quantum circuit. The baseline classifiers hyperparameters were optimized using Simultaneous Perturbation Stochastic Approximation (SPSA) optimizer and ran on a Qiskit Aer similator, which is a high-performance simulator environment for exploring quantum computing methods and applications in the noisy intermediate scale quantum (NISQ) realm.

Table 2. Performance results of quantum text classifiers

Model		Accuracy	Precision	Recall	F1-score
Spider	MC data	100%	100%	100%	100%
	RP data	37%	43%	37%	36%
Stairs	MC data	60%	60%	60%	60%
	RP data	68%	71%	68%	69%
Cups	MC data	57%	59%	57%	54%
	RP data	32%	32%	32%	32%
Tree	MC data	87%	87%	87%	87%
	RP data	63%	68%	63%	64%
DisCoCat	MC data	70%	81%	70%	67%
	RP data	58%	70%	58%	57%

According to the Table 2 the classifiers revealed varying performance across the metrics. Spider achieved excellent performance with all metrics on the MC data. However, it performance significantly dropped on the RP data. Stairs and Tree showed consistent and reasonable performance on both datasets, indicating their ability to handle diverse data while maintaining a good balance between metrics. Cups exhibited moderate performance on the MC data but struggled on the RP data. DisCoCat showed relatively high accuracy and precision but lower recall and F1-score on both datasets, indicating a tendency to be more conservative in predictions. These findings highlight the strengths and weaknesses of each classifier. In summary, all classifiers perform worse on the RP dataset over on the MC dataset, as expected, because the RP dataset is more challenging. However, DisCoCat, Stairs and Tree models perform better which explains their robustness in comparison to other models.

4 Ensemble Learning

Ensemble learning is a powerful approach to machine learning that has been successfully applied in a wide range of domains. It involves combining several base learners (individual learning algorithms) to improve the generalization ability of the learning system. An ensemble is usually more accurate and robust than any individual base classifier because it can handle different types of data and errors, as well as capture multiple aspects of the data distribution. As it has been demonstrated that ensemble learning improves model performance, the model suggested in this work is a voting ensemble model formed by five baseline quantum text classifiers: Spider, Cups, Stairs, Tree and DisCoCat. At the top level, we have used a two different voting methods in order to combine the predictions produced by the bottom level. Figure 2 illustrates the applied scheme graphically.

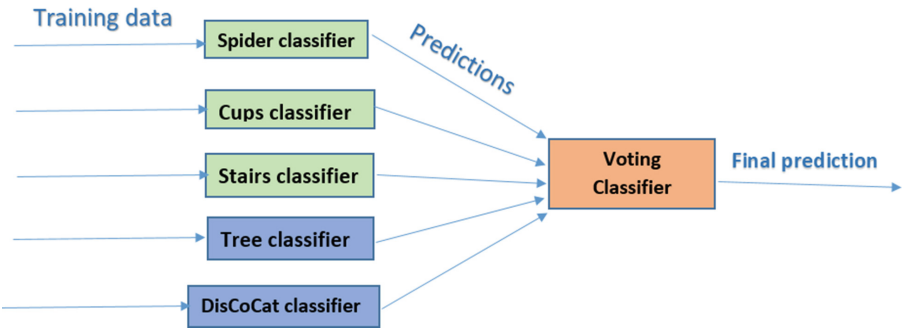


Fig. 2. A graphic illustration of the ensemble scheme utilized in the present work.

After splitting the dataset into training, validation and test data, each classifier was trained independently on the training and validation data. Then, the voting classifier feeds the test instance to each individual model and obtains its predicted class label as a distribution of probabilities. Finally, The voting classifier estimates the right label using either Hard or Soft voting methods.

In Hard Voting, the final prediction is the one with the highest frequency in the probability distributions predicted from the individual models. For example, if three models predict probability distributions of [0.1, 0.9], [0.8, 0.2] and [0.6, 0.4] then 0.9 is the highest frequency so model 1 is chosen and the result of the vote would be [0, 1] which means class two. In Soft Voting, for example, if three models predict probability distributions of [0.2, 0.8], [0.8, 0.2], and [0.6, 0.4], respectively, so the result of the soft vote would be [(0.2+ 0.8 + 0.6)/3, (0.8 + 0.2 + 0.4)/3] = [0.53, 0.46], the result of the vote would be [1, 0] which means class one.

4.1 Results and Discussion

The primary findings are presented in this section based on the results obtained on the datasets given previously running on a Qiskit Aer simulator. Using the quantum pipeline (see Fig. 1) in all the experiments, the quantum text classifiers performance are presented in Table 3.

Table 3. Performance results of Ensemble quantum text classifiers

Dataset		Accuracy	Precision	Recall	F1-score
MC dataset	Hard Voting	93%	93%	93%	93%
	Soft Voting	97%	97%	97%	97%
RP dataset	Hard Voting	70%	65%	70%	66%
	Soft Voting	68%	72%	68%	68%

According to the Table 3, in the MC dataset, both hard voting and soft voting achieved high accuracy, precision, recall, and F1-scores. Soft voting performed slightly better, with higher values of 97% across all metrics. However, on the RP dataset, the performance of both voting methods was comparatively lower. Hard voting achieved higher accuracy and recall of 70% but had lower precision and F1-score. Soft voting, on the other hand, had higher precision of 72% but lower accuracy, recall, and F1-score. Overall, the ensemble model using soft voting consistently outperformed the hard voting approach on both datasets, showcasing its effectiveness in combining the predictions of the individual classifiers.

Comparing the results of the individual classifiers with the ensemble models on both the MC and RP datasets, we observe the following trends. On the MC dataset, the ensemble models using hard voting and soft voting consistently outperformed the individual classifiers (except Spider), achieving higher accuracy, precision, recall, and F1-scores of 97%. However, on the RP dataset, the ensemble model outperform the individual classifiers, achieving higher accuracy of 70% and precision of 72%. This suggests that the ensemble approach might face challenges in effectively combining predictions or addressing the complexities specific to the RP dataset.

5 Conclusion

Lambeq was established to meet the aims of QNLP by linking the theoretical and practical aspects. Consequently, the objective of this study was to conduct a comparative evaluation of different classifiers for text classification and explored the benefits of ensemble learning. The proposed schema involves training each individual classifier using different semantic representation of sentences, on two specific QNLP datasets, MC and RP. Afterwards, a Voting classifier was formed to make predictions on test data, using two different voting methods;

hard and soft. The key findings indicate that the ensemble models, utilizing both hard voting and soft voting, consistently outperformed the individual classifiers. Accordingly, Soft voting can yield a more accurate and dependable prediction by taking the confidence of each model's prediction into consideration with. Furthermore, The ensemble approach demonstrated improved accuracy, precision, recall, and F1-scores, highlighting the effectiveness of combining the predictions from multiple classifiers. Although this study presents beneficial insights, there is opportunity for further study and improvement. Further studies could look into the usage of advanced ensemble approaches to improve classification performance. Furthermore, examining the influence of various voting schemes might contribute to improve the proposed methodology.

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